The Nelson-Aalen Estimator

2026-04-17

Introduction

The Nelson-Aalen estimator is the standard non-parametric estimator of the cumulative hazard $H(t)$ for right-censored data. It is the natural counterpart of the Kaplan-Meier estimator: where KM directly targets the survival function, Nelson-Aalen targets the cumulative hazard, and the two are linked by $S(t) = \exp(-H(t))$. Nelson-Aalen has slightly better small-sample properties than Kaplan-Meier in the right tail and is the natural input for any analysis framed in terms of hazards rather than survival probabilities, including diagnostics for proportional hazards across groups.

Prerequisites

A working understanding of cumulative hazards, the Kaplan-Meier estimator, and the relationship $S(t) = \exp(-H(t))$.

Theory

The Nelson-Aalen estimator is

$$\hat{H}(t) = \sum_{t_i \le t} \frac{d_i}{n_i},$$

with d_i the number of events at t_i and n_i the number at risk just before t_i. Each event time contributes its empirical hazard increment d_i/n_i, and the cumulative hazard is the running sum. The variance estimator is

$$\widehat{\mathrm{Var}}\big(\hat{H}(t)\big) = \sum_{t_i \le t} d_i/n_i^2,$$

which gives pointwise confidence bands on the log-log scale. The Nelson-Aalen-based survival estimator $\hat{S}(t) = \exp(-\hat{H}(t))$ differs slightly from the Kaplan-Meier product-limit estimator and is preferred when the cumulative-hazard scale is the primary reporting target.

Assumptions

Independent, non-informative censoring. Adequate at-risk sets across the follow-up window — the right tail of the estimator has wide variance when only a handful of subjects remain.

R Implementation

```r
library(survival)

data(lung)
fit <- survfit(Surv(time, status) ~ 1, data = lung)

# Cumulative hazard via NA
```

```r
plot(fit, fun = "cumhaz", main = "Nelson-Aalen cumulative hazard")

# Underlying values
summary(fit)$cumhaz |> head()
```

Output & Results

`survfit()` with `fun = "cumhaz"` plots the Nelson-Aalen cumulative hazard with pointwise confidence bands. `summary(fit)$cumhaz` returns the underlying numeric values; the corresponding event times come from `summary(fit)$time`.

Interpretation

A reporting sentence: "The Nelson-Aalen cumulative hazard reached approximately 1.5 by day 800 (95 % CI 1.3 to 1.7), corresponding to a survival probability of $\exp(-1.5) = 0.22$ at that time; the slope of $\hat{H}(t)$ in the first 200 days indicates an instantaneous hazard of approximately 0.0006 per day." Always pair the cumulative-hazard estimate with the implied survival probability for non-statistical readers.

Practical Tips

- Use Nelson-Aalen for cumulative-hazard reporting and diagnostic comparisons; use Kaplan-Meier for survival probabilities.
- Compare $\log \hat{H}(t)$ across groups: parallel lines indicate proportional hazards (the Cox-model assumption); divergent or crossing lines indicate non-proportionality.
- Construct confidence bands on the log-log scale or via $H \pm z_{\alpha/2} \cdot \mathrm{SE}(H)$; the log-log version respects the non-negativity of cumulative hazard.
- For Cox regression baseline hazard estimation, Breslow's estimator is the slight variant used internally; both reduce to Nelson-Aalen when no covariates are present.
- For very small at-risk sets at late times, the estimator's variance balloons; interpret the right tail cautiously and consider truncating the plot at the last well-sampled time.
- For competing risks, the cumulative-cause-specific hazard is the obvious analogue; the Aalen-Johansen estimator generalises Nelson-Aalen to multi-state processes.

R Packages Used

`survival` for `survfit()` with `fun = "cumhaz"`; `mstate` for multi-state extensions including the Aalen-Johansen estimator; `etm` for the empirical transition matrix; `ggsurvfit` for ggplot-flavoured Nelson-Aalen plots integrated with downstream tidyverse work.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)

- Time-dependent Brier, IPA, external validation